

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy & material cycles: natural vs human systems - Set 1

Mark scheme

AQA spec 3.6.2-3.6.3; Sustainability; textbook pp. 373-374

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels • Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures • Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x • So natural fixation is sustainable indefinitely with a low carbon footprint	4
01.2	<i>Any 3 from:</i> • Hydrological cycle (evaporation, precipitation) • Carbon cycle / photosynthesis • Atmospheric circulation / winds • Thermohaline circulation (density differences from heating / evaporation) • Decomposition	3
01.3	• Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting • Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) • Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	

Question	Marking guidance	Marks
02.1	- Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates - Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials - Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult - Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable	4
02.2	- Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life - They leach from landfill or are released by incineration; persistent, bioaccumulating	2
02.3	- Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) - Easy identification of materials (coding) and separation of components so they can be recycled - Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) - Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
03.1	<i>Any 2 from:</i> - Hydrological cycle - Carbon cycle - Nitrogen cycle - Atmospheric circulation - Thermohaline circulation - Photosynthesis	2
03.2	- Enzymes reduce the activation energy of reactions - So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures	2
03.3	<i>Any 1 from:</i> - Haber process (nitrogen fixation) - Incineration of wastes (decomposition) - Many manufacturing processes	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
04.1	- Energy: low energy-density renewable (solar) / non-renewable fossil fuels - Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion - Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate - Temperature: low / high (Haber process, incineration)	4

Question	Marking guidance	Marks
04.2	- Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) - They are re-used in cycles so there is no depletion of raw materials or build-up of wastes	2
04.3	- Use often disperses the minerals (eg in products spread over the world / corrosion) - Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced	2
04.4	- They cannot easily be removed from end-of-life appliances and are not biodegradable - Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
05.1	Waste products that are the raw materials for other processes	1
05.2	- Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) - The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2
05.3	- Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials - Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	